WENMIAO JIA

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Website: wenmiaojia.github.io/personal-website
Computer Engineering @ UW–Madison | GPA 4.00 | Expected May 2028

EXPERIENCE

Undergraduate Researcher Apr 2026 – Present
UW Connect Lab, UW–Madison Madison, WI

- **ECE SPARK Summer Research Fellow** (inaugural cohort, 2026) — full-time, 10-week funded appointment advised by Prof. Parmesh Ramanathan and Prof. Bhuvana Krishnaswamy. Continuing research for academic credit.
- Built a GNSS simulation pipeline from SignalSim (transmitter) to GNSS-SDR (receiver) to positioning modeling (RTKLIB) and then analyzed the result in MATLAB
- Injected a known tropospheric delay and measured the recovery error validated at both 1.5 GHz and 12 GHz
- Recovered zenith tropospheric delay to 2 cm from the injected truth; swept the delay values injected to confirm the recovery is linear and cross verified against EarthScope/GAGE station
- Implemented dual-frequency L1/L2 to enable the iono-free combination and quantified its noise penalty: ±255 mm vs ±52 mm single-frequency

Firmware Team Member Sep 2025 - Present
Badger Solar Racing Madison, WI

- Implemented a dashboard in the car with real-time speedometer, turning signals, gear shifts and headlight
- Decoded and encoded CAN messages to process the vehicle telemetry to display the state of each component
- Implemented cruise control in the car to optimize energy efficiency during optimum speeds

EDUCATION

Bachelor of Science in Computer Engineering Sep 2025 - May 2028
University of Wisconsin-Madison | GPA: 4.00 Madison, WI

PROJECTS

RP2040 Keypad Door Lock | Custom PCB | *Python*, *Raspberry Pi Pico*, *PCB-Design*

- Built a keypad-entry door lock on the RP2040 (Raspberry Pi Pico) with password validation and servo actuation
- Designed a custom PCB integrating an LCD, custom numpad, and servo driver

Engineering Hall Navigation App | App Development | *Kotlin*, *Android Studio*, *Algorithms*

- Rebuilt an Android app for indoor Engineering Hall navigation with a custom interface modeled on Google Maps
- Implemented a pathfinding function over a node graph using breadth-first search

ACHIEVEMENTS

- A-Level Further Mathematics High Achievement
- "Atmospheric Sensing Using Global Navigation Satellite Systems," poster, ECE SPARK Symposium, UW–Madison, Summer 2026 (first author)
- Tau Beta Pi Engineering Honor Society, Wisconsin Alpha Chapter, University of Wisconsin–Madison
- Paul H. Thiele Scholarship, University of Wisconsin–Madison College of Engineering (2026)
- Dean's List, University of Wisconsin–Madison College of Engineering (Fall 2025, Spring 2026)

SKILLS

Languages: Python, C, Java, C++, HTML, MATLAB, Assembly, Kotlin
Frameworks/Platforms: Linux, GNSS-SDR, gps-sdr-sim, Android Studio, PyTorch, Bash scripting, GitHub
Signal Processing: GNSS: SDR signal processing, GNSS signal simulation, PPP, ZTD estimation